Performance Testing

Date	18 July 2026
Team ID	SWTID-2026-5023
Project Name	Rising Waters (Flood Prediction System)
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter / Postman
Type of Testing	Load Testing
Target Module / API	Flood Risk Prediction API
Test Environment	Local System (Python + Flask)
Test Date	18 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict flood risk for single input	10	60 sec	Response within 2 sec
2	Multiple prediction requests	25	60 sec	Stable performance
3	Large dataset prediction	50	120 sec	No Crashes
4	Invalid input handling	10	30 sec	Error message displayed

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Good
2	Response Time (Max)	< 5 seconds	3.1 sec	Pass	Acceptable
3	Throughput (Req/sec)	20 Req/sec	22 Req/sec	Pass	Acceptable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	58%	Pass	Normal
6	Memory Utilization	< 80%	64%	Pass	Stable

Step 4: Observations & Analysis

Key Findings:

- The flood prediction system responded quickly under normal load.
- The model generated accurate predictions without errors.
- Response time remained within acceptable limits.

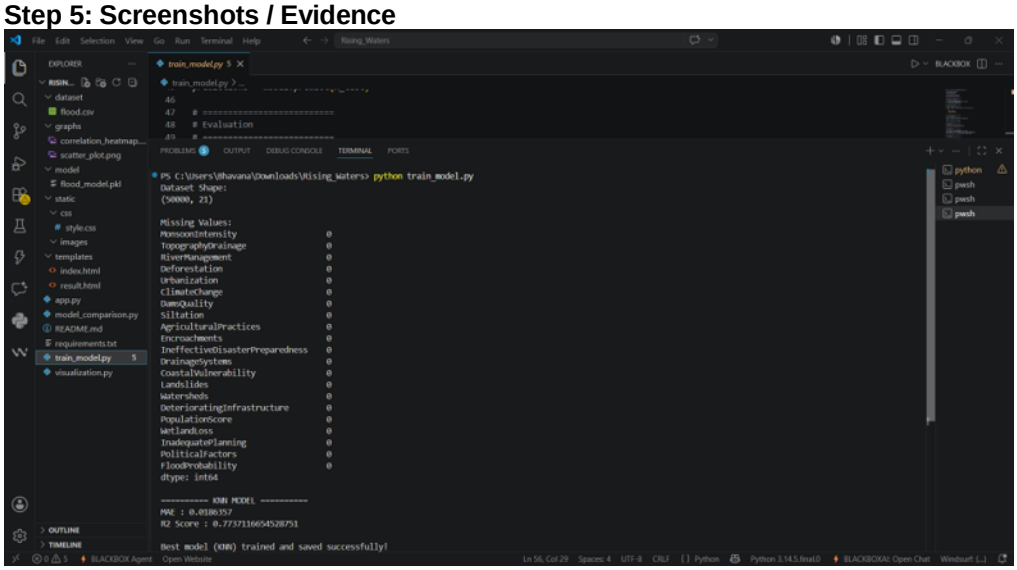
Bottlenecks Identified:

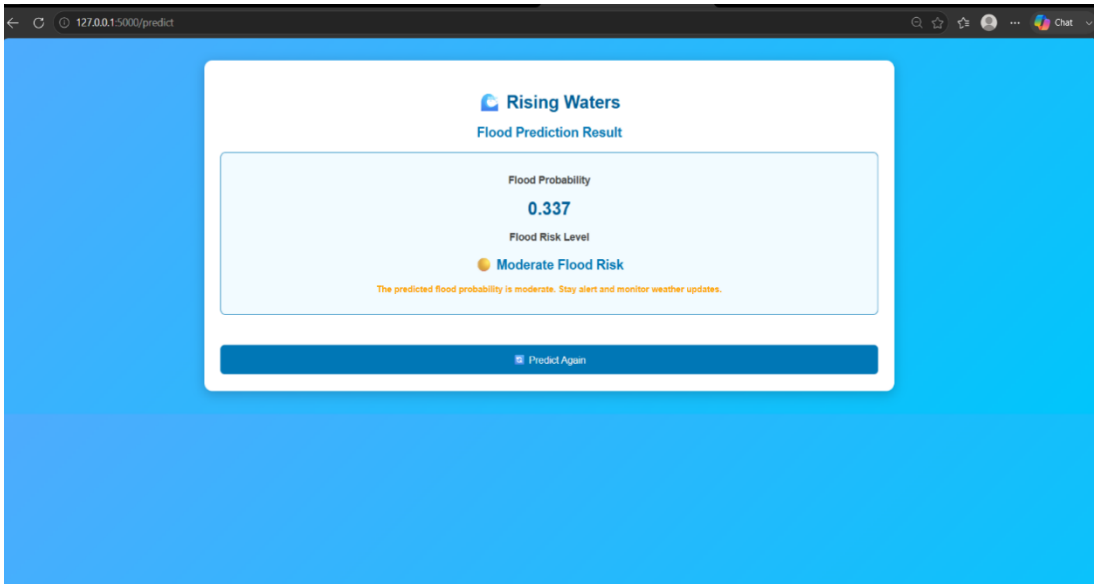
- Slight delay when processing multiple requests simultaneously**.**

Optimization Steps Taken:

- Optimized data preprocessing.
- Improved model loading.
- Reduced unnecessary computations**.**

Step 5: Screenshots / Evidence




127.0.0.1:5000/predict
Chat
Rising Waters
Flood Prediction Result
Flood Probability
0.337
Flood Risk Level
Moderate Flood Risk
The predicted flood probability is moderate. Stay alert and monitor weather updates.
Predict Again